

One Wrong Move: Advanced Pattern Actions Report

Report 008 · Branch `advanced-pattern-actions`

Summary

Implemented the next major puzzle-design upgrade: the model now supports single-answer, best-move, and multi-select puzzles. Rule Rows, Conveyor Shift, and Knight Path are retired from daily play. Chess Attack was rebuilt around numbered pieces on the same square. Go became a real source world with capture-max and liberty-selection puzzles. The daily selector now avoids retired types, prefers source-world variety, and keeps lower-is-better scoring intact.

Repo State Summary

- Working directory: `/Users/kevinhegg/Desktop/one-wrong-move`
- Starting branch: `main` ; created branch: `advanced-pattern-actions`
- Remote: `git@github.com:KevinHegg/one-wrong-move.git`
- No nested repository found beyond `./.git`.
- Local server verified on `http://127.0.0.1:8080/`.

Git Commands Run

```
pwd, git status --short, git branch --show-current, git log --oneline -8, git remote -v, find . -maxdepth 3 -name .git -type d, ls -la, git switch -c advanced-pattern-actions || git switch advanced-pattern-actions.
```

Files Created Or Changed

- `README.md`
- `docs/game-design.md`
- `docs/pattern-systems.md`
- `index.html`
- `public/index.html`
- `public/style.css`
- `public/game.js`
- `public/puzzles.js`
- `public/lab.html`
- `public/lab.js`
- `scripts/validate-puzzles.js`
- `scripts/validate-pattern-systems.js`

- docs/codex-reports/screenshots/008-*.png
- docs/codex-reports/008-advanced-pattern-actions-report.pdf

Puzzles Retired From Daily Play

- **Rule Rows:** too easy to solve by diagonal or missing-symbol scanning.
- **Conveyor Shift:** too easy to solve by phase scanning instead of reasoning.
- **Knight Path:** valid but narrow; Chess Attack gives chess movement more variety and texture.

New Answer Modes

- **identifyOne** : tap the one rule-breaking symbol.
- **chooseOne** : tap the one best move square.
- **multiSelect** : select the exact answer set and press Submit.

New And Updated Puzzle Types

New or substantially upgraded types include Chess Attack, Go Capture Max, Go Liberties, Poker Hand Trap, Train Route, Card Straight, Suit Cycle, Logic Gate Row, Domino Chain, Dice Sum, Mirror Trap, and Pair Pact.

Type	Source world	Answer mode	Difficulty	Rule
Suit Cycle	Cards	identifyOne	1	Suits cycle across rows and shift by row.
Pair Pact	Relationships	identifyOne	1	Established real-world partner pairs.
Domino Chain	Dominoes	identifyOne	1	Neighboring domino halves must match.
Dice Sum	Dice	identifyOne	1	Dice rows meet small target sums.
Card Straight	Cards	identifyOne	2	Rank progressions with suit behavior.
Logic Gate Row	Logic	identifyOne	2	0/1 inputs and gate outputs.
Mirror Trap	Relationships	identifyOne	2	Mirrored side transforms into paired counterparts.
Rotation Logic	Compass	identifyOne	2	Directions rotate by a fixed step.
Latin Trap	Symbol Grammar	identifyOne	2	Rows and columns contain one of each symbol.
Animal Food Web	Ecology	identifyOne	2	Food-chain order must stay coherent.
Compass Rose	Compass	identifyOne	2	Bearings follow a rotation sequence.
Chess Attack	Chess	identifyOne	3	Numbered piece must attack the next numbered piece.
Poker Hand Trap	Cards	identifyOne	3	Rows declare pair, flush, run, or trips.
Train Route	Routes	identifyOne	3	Track tiles form one continuous route from S to F.
Checkers Jump	Checkers	identifyOne	3	Numbered checkers move diagonally.
Go Capture Max	Go	chooseOne	4	Choose the move that captures the most white stones.
Go Liberties	Go	multiSelect	4	Select every liberty of a marked group.

Chess Attack Design Details

Each occupied square renders a chess piece with its sequence number in a small badge. The player follows the numbers; each piece must legally attack the next numbered piece. The validator computes rook, bishop, queen, knight, and king attacks directly, excludes pawns, and identifies the first illegal attacker.

Go Puzzle Design Details

Go uses simplified 5x5 rules: orthogonal adjacency connects groups, empty orthogonal points are liberties, and diagonal points do not count. Go Capture Max asks for the one move that captures the most white stones. Go Liberties asks the player to select every liberty of a marked group.

Daily Selection Rules

The daily mix still has three deterministic rounds. It excludes retired types, prefers three different source worlds, avoids duplicate card/Go/movement families when possible, and escalates from easier round 1 to strongest round 3.

Candidate Pattern Systems

Candidate	Source world	Symbol set	Rule mechanic	Break modes	Difficulty	Status
Rule Rows	Abstract grammar	Moon/crown/diamond/eye	Shifted row recipe	Wrong next, duplicate, outside symbol	1	Retired/lab
Conveyor Shift	Abstract grammar	Five glyph tiles	Row-to-row shift	Wrong position, duplicate, neighbor column	2	Retired/lab
Rotation Logic	Compass	N/E/S/W/diagonals	Fixed rotation step	Wrong turn, opposite, skipped bearing	2	Implemented
Latin Trap	Latin square	Five glyphs	One of each per row/column	Row duplicate, column duplicate	2	Implemented
Pair Pact	Relationships	Moon/sun, key/lock, bee/flower	Stable partner pairs	Wrong partner, repeated leader	1	Implemented
Mirror Trap	Mirror mapping	Source/copy symbol pairs	Mirror plus transformation	Unchanged, wrong pair, swapped partner	2	Implemented
Card Straight	Playing cards	Ranks and suits	Rank progression with suit rhythm	Wrong rank, wrong suit, duplicate	2	Implemented
Suit Cycle	Playing cards	Ranks and suits	Suit cycle shifts by row	Wrong cycle position, duplicate suit	1	Implemented
Poker Hand Trap	Playing cards	Hand labels and cards	Rows form named hand classes	Broken pair, flush, run, trips	3	Implemented
Knight Path	Chess	Numbered knights	Legal L-moves	Orthogonal, diagonal, wrong L	3	Retired/lab

Candidate	Source world	Symbol set	Rule mechanic	Break modes	Difficulty	Status
Chess Attack	Chess	King/queen/rook/bishop/knight	Piece attacks next piece	Illegal rook, bishop, queen, knight, king	3	Implemented
Checkmate Net	Chess	Kings, guards, targets	Target squares controlled	Uncovered escape, blocked line	5	Backlog
Checkers Jump	Checkers	Numbered checkers	Diagonal movement	Orthogonal, wrong diagonal, fake jump	3	Implemented
Go Capture Max	Go	Black/white stones	Best capture move	Tie trap, non-capture, wrong liberty	4	Implemented
Go Liberties	Go	Marked group and empties	Select all liberties	Missing liberty, diagonal false liberty	4	Implemented
Go Save Group	Go	Atari groups	Choose saving or capture move	Self-atari, false capture	4	Backlog
Logic Gate Row	Digital logic	0/1 gates	Truth-row output	Wrong AND, OR, XOR, NAND	2	Implemented
Circuit Trace	Electronics	Wires/gates/outputs	Signals flow through circuit	Broken wire, inverted output	4	Backlog
Domino Chain	Dominoes	Numeric halves	Neighbor halves match	Left mismatch, right mismatch	1	Implemented
Dice Sum	Dice	Die faces and targets	Small arithmetic totals	Sum high, sum low, wrong die	1	Implemented
Train Route	Transit maps	S/F/track glyphs	Continuous route	Broken turn, dead end	3	Implemented
Animal Food Web	Ecology	Plant/insect/frog/snake/hawk	Food-chain order	Wrong habitat, wrong predator/prey	2	Implemented
Compass Rose	Compass/clock	Bearings	Repeated rotation	Wrong amount, opposite	2	Implemented
Calendar Week	Calendar	Days/weekend markers	Day cycle and grouping	Skipped day, wrong weekend	2	Backlog
Music Measure	Music	Notes/rests/beats	Equal beat totals	Too many/few beats	3	Backlog
Chemistry Valence	Chemistry	Atoms/bonds	Bond counts match valence	Overbonded, underbonded	4	Backlog
Sports Formation	Sports	Positions/lanes/ball	Coverage formation	Uncovered lane, illegal receiver	3	Backlog
Language Pattern	Linguistics	Prefix/root/suffix	Grammar slot pattern	Bad suffix, agreement error	4	Backlog
Map Compass Walk	Maps	Directions/landmarks	Bearings to landmarks	Wrong turn, missed landmark	3	Backlog
Set Attribute Grid	Set-style logic	Shape/fill/count/orientation	Attribute rows	Wrong fill, count, orientation	4	Backlog

Validation Results

- `node --check public/symbols.js` passed.

- `node --check public/scoring.js` passed.
- `node --check public/puzzles.js` passed.
- `node --check public/game.js` passed.
- `node --check public/lab.js` passed.
- `node scripts/validate-scoring.js` : validated lower-is-better scoring cases.
- `node scripts/validate-puzzles.js` : validated 4,500 daily rounds across 300 dates and 20 puzzle types.
- `node scripts/validate-pattern-systems.js` : validated 20 total types, 17 active types, 300 dates, and 5 attempts.
- `git diff --check` passed.
- Headless browser completion flow reached `complete` , puzzle validation was `true` , and browser console errors were 0.

Local Run Instructions

Open `public/index.html` directly, or run:

```
python3 -m http.server 8080 -d public
```

Then visit `http://localhost:8080` . The lab is at `http://localhost:8080/lab.html` .

Netlify Deploy Settings

- Build command: leave blank.
- Publish directory: `public` .

Screenshot Captures

All required screenshots were captured automatically and saved in `docs/codex-reports/screenshots/` with `008-` filenames.

Intro screen

PUZZLE #001

One Wrong Move

TIME

0.0s

READY

DAILY PUZZLE

MISTAKES 0

DAILY PUZZLE READY

One board. One rule. One move — or one precise set of moves.

Score = rounded-up solving time + 10s per mistake.
Lower is better.

The board stays hidden until you open a round briefing.

Start

Single-answer briefing

PUZZLE #001

One Wrong Move

TIME

0.0s

ROUND 1 OF 3

DAILY PUZZLE

MISTAKES 0

ROUND 1: PAIR PACT · RELATIONSHIPS

**Symbols travel in real-world partner pairs.
Tap the symbol with the wrong partner.**

Study the mechanic. The board stays hidden and the timer is paused.

SYMBOLS:



EXAMPLE



Moon pairs with sun; key pairs with lock.

Round 1 is hidden until you press Start
Round.

Start Round

Restart

Round 1 active board

PUZZLE #001

One Wrong Move

TIME

0.2s

ROUND 1 OF 3

DAILY PUZZLE

MISTAKES 0

ROUND 1: PAIR PACT · RELATIONSHIPS

**Symbols travel in real-world partner pairs.
Tap the symbol with the wrong partner.**

Infer the rule from the board, then tap the one symbol that breaks it.

				
BE	FL			
				
				
BE	FL			

Restart

Wrong tap state

PUZZLE #001

One Wrong Move

TIME

0.6s

ROUND 1 OF 3

DAILY PUZZLE

MISTAKES 1

ROUND 1: PAIR PACT · RELATIONSHIPS

**Symbols travel in real-world partner pairs.
Tap the symbol with the wrong partner.**

Infer the rule from the board, then tap the one symbol that breaks it.

Try again

Look for pairs that repeat elsewhere; one partner is with the wrong mate.

				
BE	FL			
				
				
BE	FL			

Restart

Correct feedback state

PUZZLE #001

One Wrong Move

TIME

0.7s

ROUND 1 OF 3

DAILY PUZZLE

MISTAKES 1

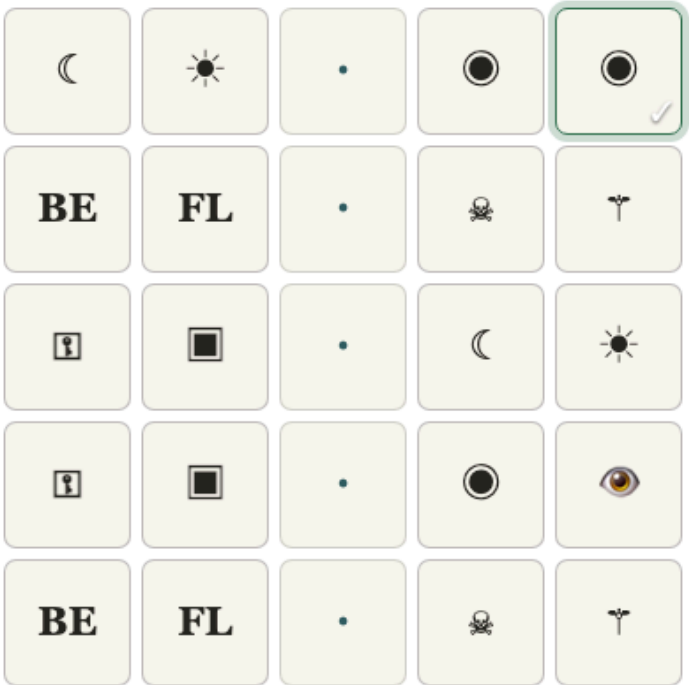
ROUND 1: PAIR PACT · RELATIONSHIPS

**Symbols travel in real-world partner pairs.
Tap the symbol with the wrong partner.**

Infer the rule from the board, then tap the one symbol that breaks it.

Correct

That symbol breaks the pairing pact.



Next Round

Restart

Logic Gate Row active board

PUZZLE #001

One Wrong Move

TIME

0.9s

ROUND 2 OF 3

DAILY PUZZLE

MISTAKES 1

ROUND 2: LOGIC GATE ROW · LOGIC

Each row is A, B, gate, equals, output. Tap the wrong output.

Infer the rule from the board, then tap the one symbol that breaks it.

				
BE	FL			
				
				
BE	FL			

Restart

Multi-select briefing

PUZZLE #001

One Wrong Move

TIME

1.0s

ROUND 3 OF 3

DAILY PUZZLE

MISTAKES 1

ROUND 3: GO LIBERTIES · GO

Tap every liberty of the marked group, then submit.

This round uses multiple selections. The board stays hidden and the timer is paused.

SYMBOLS:



EXAMPLE



Only empty orthogonal neighbors are liberties; diagonals do not count.

Round 3 is hidden until you press Start Round.

Start Round

Restart

Go Liberties active board

PUZZLE #001

One Wrong Move

TIME

1.2s

ROUND 3 OF 3

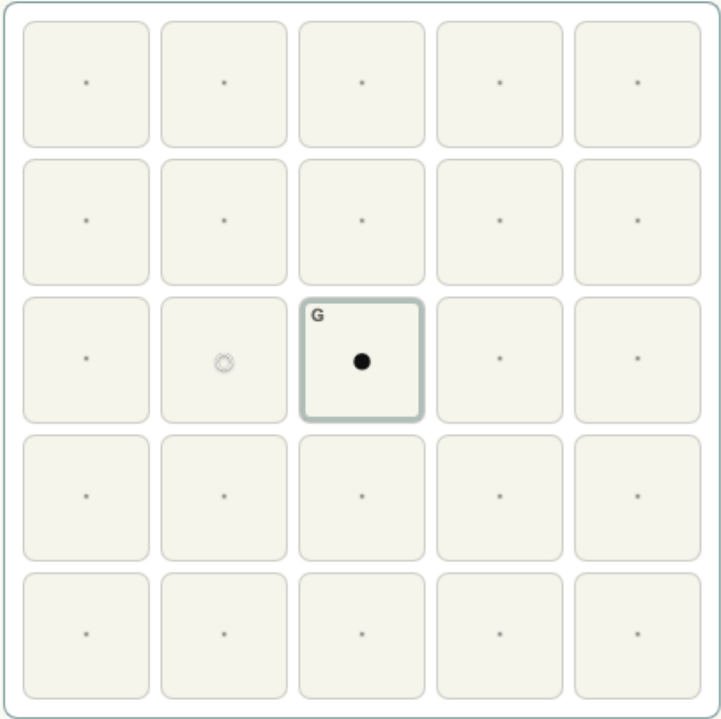
DAILY PUZZLE

MISTAKES 1

ROUND 3: GO LIBERTIES · GO

Tap every liberty of the marked group, then submit.

Select every matching square, then submit.



Submit liberties

Clear selection

Restart

Go Liberties selected liberty

PUZZLE #001

TIME

One Wrong Move

1.5s

ROUND 3 OF 3

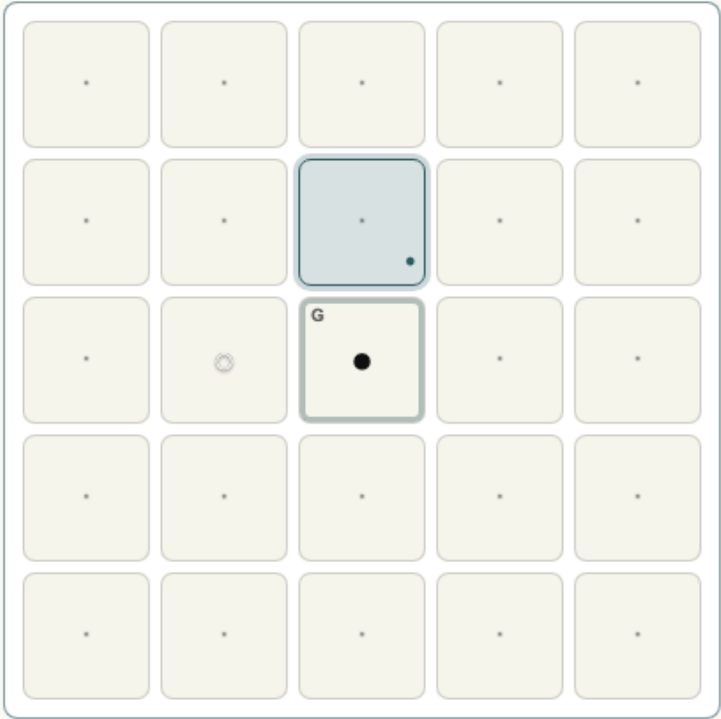
DAILY PUZZLE

MISTAKES 1

ROUND 3: GO LIBERTIES · GO

Tap every liberty of the marked group, then submit.

Select every matching square, then submit.



Submit liberties

Clear selection

Restart

Wrong multi-select submission

PUZZLE #001

One Wrong Move

TIME

1.9s

ROUND 3 OF 3

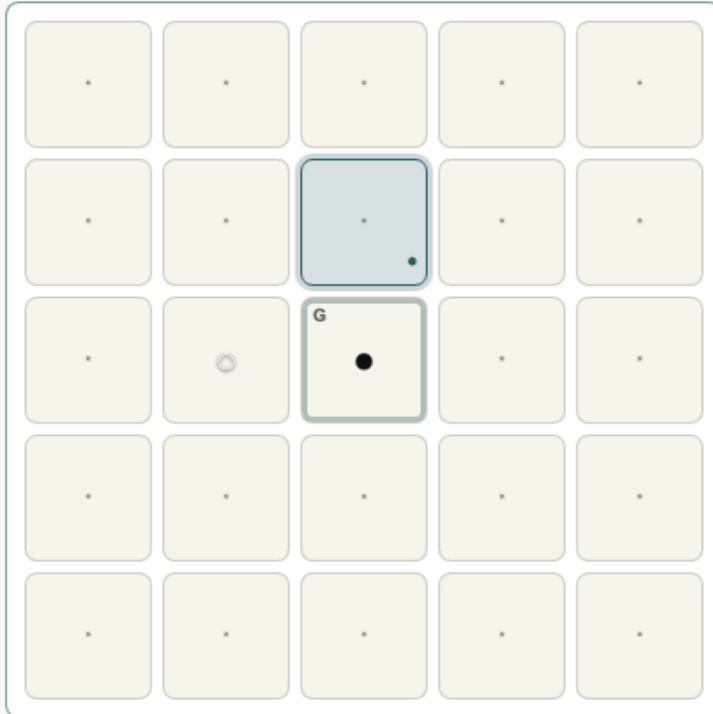
DAILY PUZZLE

MISTAKES 1

ROUND 3: GO LIBERTIES · GO

Tap every liberty of the marked group, then submit.

Select every matching square, then submit.



Submit liberties

Clear selection

Restart

Completion/share screen

PUZZLE #001

One Wrong Move

TIME

2.6s

COMPLETE

DAILY PUZZLE

MISTAKES 1

DAILY PUZZLE SOLVED

Solved 3 of 3

Score = rounded-up solving time + 10s per mistake.
Lower is better.

SOLVED

3 of 3

ACTIVE TIME

2.6s

BASE TIME

3s

FINAL SCORE

13s

LOWER IS BETTER

Mistakes **1**

Penalty **$1 \times 10s = +10s$**

Formula **$3s + 10s = 13s$**

SHARE TEXT

One Wrong Move #001

Score: 13s

Lower is better

Base: 3s

Mistakes: $1 \times 10s = +10s$



Share result

Play another mix

Restart

Variant replay screen

PUZZLE #001

One Wrong Move

TIME

0.0s

READY

VARIANT 2

MISTAKES 0

REPLAY VARIANT READY

One board. One rule. One move — or one precise set of moves.

Score = rounded-up solving time + 10s per mistake.
Lower is better.

The board stays hidden until you open a round briefing.

Start Variant

Puzzle lab overview

DEVELOPMENT LAB

One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

Regenerate

Production puzzle types (17)

CARDS · DIFFICULTY 1 · IDENTIFYONE

Show answer

Suit Cycle

Suits cycle across the row and shift by row. Tap the card with the wrong suit.

A♠2♥3♦4♣7♥9♠J♣Q♦

A♥	A♦	A♣	A♠	A♥
3♦	3♠	3♣	3♥	3♦
5♣	5♠	5♥	5♦	5♣
7♣	7♥	7♦	7♣	7♠
9♥	9♠	9♣	9♠	9♥

Break signature

suit-cycle|duplicate-suit|6|3*3{"rank":"3","suit":"diamond","suitIndex":1}

Evidence

The rows shift a four-suit cycle; the tapped card uses the wrong suit for its position.

RELATIONSHIPS · DIFFICULTY 1 · IDENTIFYONE

Show answer

Pair Pact

Symbols travel in real-world partner pairs. Tap the symbol with the wrong partner.

☾☀☷☴☳☶BEFL

☼	☶	☷	☵	☴
☾	☀	☷	☵	☴
☼	☶	☷	BE	FL
☼	☶	☷	☼	☵
BE	FL	☷	☾	☾

Break signature

pair-pact|leader-repeated|24|<"☾"

Evidence

The board repeats real-world pair pacts; the tapped symbol is not the established partner.

DOMINOES · DIFFICULTY 1 · IDENTIFYONE

Show answer

Domino Chain

Neighboring domino halves must match. Tap the domino that breaks the chain.

123456

4 5	5 6	6 1	1 2	2 3
5 6	6 1	1 2	2 3	3 4
6 1	1 2	2 3	3 4	4 5
1 2	2 3	3 4	4 2	5 6
2 3	3 4	4 5	5 6	6 1

Break signature

domino-chain|right-side-mismatch|18|4|2|"left":4,"right":2}

Evidence

Neighboring domino halves match across every clean row; the tapped domino breaks that handoff.

DEVELOPMENT LAB

One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

Regenerate

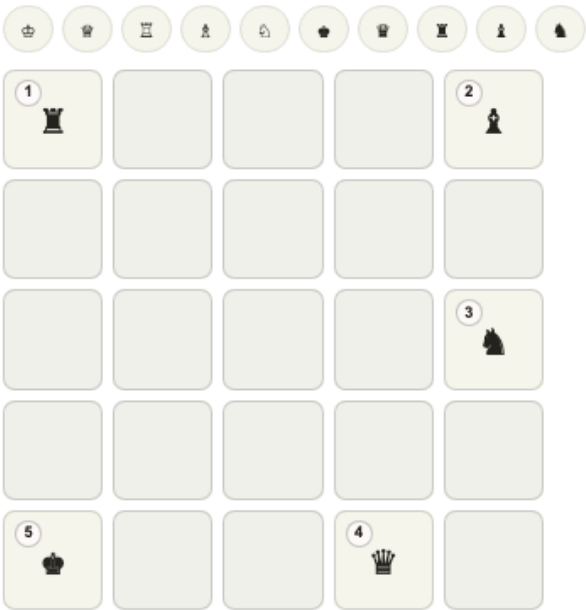
Production puzzle types (1)

CHESS · DIFFICULTY 3 · IDENTIFYONE

Show answer

Chess Attack

Follow the numbers. Each chess piece must attack the next numbered piece. Tap the first illegal attacker.



Break signature

chess-attack|bishop-file-fail|4|piece-2

Evidence

Number 2 is the first piece whose chess attack does not reach the next number.

Answer

Hidden

DEVELOPMENT LAB

One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

Regenerate

Production puzzle types (1)

CHESS · DIFFICULTY 3 · IDENTIFYONE

Hide answer

Chess Attack

Follow the numbers. Each chess piece must attack the next numbered piece. Tap the first illegal attacker.



Break signature

chess-attack|bishop-file-fail|4|piece-2

Evidence

Number 2 is the first piece whose chess attack does not reach the next number.

Answer

Cell 5: That numbered piece is the first one that cannot attack the next piece.

Card Straight active sample

DEVELOPMENT LAB

One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

Regenerate

Production puzzle types (1)

CARDS · DIFFICULTY 2 · IDENTIFYONE

Card Straight

Show answer

Each row is a short straight. Tap the card that breaks rank or suit logic.

A♠2♥3♦4♠7♥9♠J♠Q♦

5♥5♥	6♦6♦	7♣7♣	8♠8♠	9♥9♥
6♦6♦	7♣7♣	8♠8♠	9♥9♥	10♦10♦
7♣7♣	8♠8♠	9♥9♥	Q♦Q♦	J♠J♠
8♠8♠	9♥9♥	10♦10♦	J♠J♠	Q♠Q♠
9♥9♥	10♦10♦	J♠J♠	Q♠Q♠	K♥K♥

Break signature

card-straight|wrong-rank|13|Q♦Q{"rank":"Q","suit":"diamond","suitIndex":1}

Evidence

Each row forms a five-card straight with a consistent suit, so the tapped card breaks rank or suit logic.

DEVELOPMENT LAB

One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

Regenerate

Production puzzle types (1)

CARDS · DIFFICULTY 1 · IDENTIFYONE

Suit Cycle

Show answer

Suits cycle across the row and shift by row. Tap the card with the wrong suit.

A♠2♥3♦4♣7♥9♠J♠Q♦

A A♥	A A♦	A A♣	A A♠	A A♥
3 3♦	3 3♦	3 3♠	3 3♥	3 3♦
5 5♣	5 5♠	5 5♥	5 5♦	5 5♣
7 7♠	7 7♥	7 7♦	7 7♣	7 7♠
9 9♥	9 9♦	9 9♣	9 9♠	9 9♥

Break signature

suit-cycle|duplicate-suit|6|3♦3{"rank":"3","suit":"diamond","suitIndex":1}

Evidence

The row shifts a four-suit cycle, the top row contains the wrong suit for the position

DEVELOPMENT LAB

One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

Regenerate

Production puzzle types (1)

CHESS · DIFFICULTY 3 · IDENTIFYONE

Chess Attack

Show answer

Follow the numbers. Each chess piece must attack the next numbered piece. Tap the first illegal attacker.



Break signature

chess-attack|bishop-file-fail|4|piece-2

DEVELOPMENT LAB

One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

Regenerate

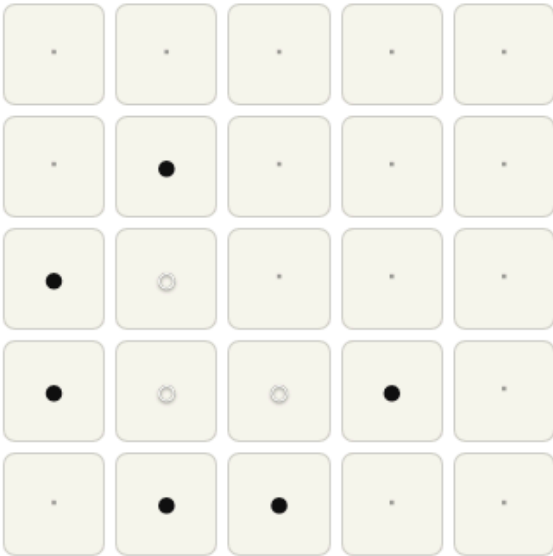
Production puzzle types (1)

GO · DIFFICULTY 4 · CHOOSEONE

Go Capture Max

Show answer

Black to play. Tap the empty intersection that captures the most white stones.



Break signature

go-capture-max|capture-three-rotate180|12|captures-3

Evidence

The board is a 5x5 grid. Black stones are at (2,2), (3,1), (4,1), and (5,2). White stones are at (3,2), (4,2), and (5,1). Empty intersections are at (1,1), (1,2), (1,3), (1,4), (1,5), (2,1), (2,3), (2,4), (2,5), (3,3), (3,4), (3,5), (4,3), (4,4), (4,5), (5,3), (5,4), and (5,5).

DEVELOPMENT LAB

One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

Regenerate

Production puzzle types (1)

LOGIC · DIFFICULTY 2 · IDENTIFYONE

Logic Gate Row

Show answer

Each row is A, B, gate, equals, output. Tap the wrong output.

0	1	AND	OR	XOR	NOT	NAND	=
1	0	AND	=	0			
0	0	OR	=	0			
1	1	XOR	=	1			
0	1	NAND	=	1			
1	0	AND	=	0			

Break signature

logic-gate-row|xor-output-wrong|14|1"1"

Evidence

Row 1: 1, 0, XOR, 1. The output is 1, which is correct. Row 2: 0, 0, OR, 0. The output is 0, which is correct. Row 3: 1, 1, XOR, 1. The output is 1, which is correct. Row 4: 0, 1, NAND, 1. The output is 1, which is correct. Row 5: 1, 0, AND, 0. The output is 0, which is correct. Row 6: 0, 1, AND, 0. The output is 0, which is correct. Row 7: 1, 1, AND, 1. The output is 1, which is correct. Row 8: 0, 0, AND, 0. The output is 0, which is correct. Row 9: 1, 0, OR, 1. The output is 1, which is correct. Row 10: 0, 1, OR, 1. The output is 1, which is correct. Row 11: 1, 1, OR, 1. The output is 1, which is correct. Row 12: 0, 0, OR, 0. The output is 0, which is correct. Row 13: 1, 0, XOR, 1. The output is 1, which is correct. Row 14: 0, 1, XOR, 1. The output is 1, which is correct. Row 15: 1, 1, XOR, 0. The output is 0, which is wrong.

DEVELOPMENT LAB

One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

Regenerate

Production puzzle types (1)

DOMINOES · DIFFICULTY 1 · IDENTIFYONE

Domino Chain

Show answer

Neighboring domino halves must match. Tap the domino that breaks the chain.

1 2 3 4 5 6

2 3	3 4	4 5	5 6	6 1
3 4	4 5	5 6	6 1	1 2
4 5	5 6	6 1	1 2	2 3
5 6	6 1	1 2	2 6	3 4
6 1	1 2	2 3	3 4	4 5

Break signature

domino-chain|right-side-mismatch|18|2|6{"left":2,"right":6}

Evidence

Neighboring domino halves must match across every chain row. The second domino breaks the chain as it has a 6 on its right side, while the first domino has a 2 on its right side.

Train Route active sample

DEVELOPMENT LAB

One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

Regenerate

Production puzzle types (1)

ROUTES · DIFFICULTY 3 · IDENTIFYONE

Train Route

Show answer

Track tiles should form one continuous route from S to F. Tap the broken tile.

S F — | r 7 L J

S	—	—	7	
			—	
			L	7
				F

Break signature

train-route|wrong-straight|8|—"—"

Evidence

The route has one continuous connection from S to F, the broken tile is not the track, the wrong way